



COSMONAUTS

On March 18, 1965, cosmonaut Alexei Leonov exited Voskhod 2 to perform the first space walk (right). He was protected by his

pressurized space suit and his eyes were shielded from the Sun's glare by mirrored goggles worn inside his helmet. Soviet manned missions continued after the Voskhod flights with the Soyuz spacecraft (top).

landing in a forest 2,000 miles (3,200km) from the target, Leonov had become the first man to walk in space.

America's first space walk was a different experience to that which the cosmonauts had experienced. Astronaut Edward H. White floated out of the main hatch of his Gemini 4 spacecraft on June 3, 1965. Unlike Leonov, White had a handheld device that enabled him to move around at will. It consisted of two small tanks of compressed gas which, released in small bursts, produced enough thrust to propel the astronaut around. He was, of course, attached to the spacecraft by an umbilical line. White was so euphoric at his weight-free acrobatics that he jokingly threatened to refuse to come back in at the end of his planned 10 minutes outside.

White and fellow crewman James McDivitt stayed in space for four days. This was also a major landmark. It was longer than the total of all previous US space flights added together. Although the Soviets had made longer flights, there were fears that the astronauts might suffer serious ill-effects on returning from such a time in zero-gravity conditions. When White and McDivitt were retrieved from splashdown evidently fit and happy, another possible barrier to long-distance space flight had evaporated.

sending two astronauts into space on board a Gemini spacecraft, Korolëv went one better. In October 1964 he put three men into orbit in a craft called Voskhod. It was a desperately risky venture. Voskhod was no more than an improved version of the old Vostok spacecraft. The only way to fit three people inside was to have them fly without space suits and to strip out the ejection seats. But Boris Yegarov, Vladimir Komorov, and design engineer Konstantin Feoktistov made a successful day trip into space and landed inside the capsule.

Because of Soviet secrecy, the Americans had no idea of the desperate straits to which Korolëv had been driven in order to carry off the Voskhod 1 mission. Nikita Khrushchëv was deposed as Soviet leader on October 12, 1964, not long after Voskhod 1 was launched, although the reasons had nothing to do with space. His replacements, the stolid Leonid Brezhnev and Alexei Kosygin, were much less reliable supporters of an extravagant space program than the ebullient Khrushchëv. In the United States, by contrast, the death of President Kennedy in 1963 had only brought the equally space-minded Lyndon B. Johnson to the White House. While the American space effort was concentrated on a single politically determined goal – putting a man on the Moon – the Soviet program increasingly lacked direction. Yet with less resources at their disposal, the Soviets could ill-afford to disperse them.

The last moment when the Soviets appeared seriously to be ahead of NASA was with the first space walk during the Voskhod 2 mission in March 1965. Yet this again was a risky improvisation rather than a carefully considered step in a planned program. The death of Korolëv in January 1966 – which brought his instant transformation from an anonymous engineer to a publicly hailed Soviet hero – was another important setback. With their experience of powerful rockets and of space vehicles, the Soviets were in no way out of contention in the race to the Moon, but by 1965–6 the United States was leading the race.

In preparation for the Gemini and Apollo programs, NASA took in new recruits to the astronaut corps. This time there were few hesitations about exchanging the status of test pilot for that of astronaut. Being fired into space with limited control of the craft had not seemed an attractive option to many men used to piloting the latest jet fighters or X-planes. Now there was not only the chance of setting foot on the Moon, but also manning spacecraft that could genuinely be controlled in space by their pilots.

Gemini missions

The first manned Gemini mission, Gemini 3, blasted off on March 23, 1965. The crew consisted of Gus Grissom from the original Mercury 7 and rookie John Young. The craft they journeyed in was made up of two parts: the spacecraft proper and a support module containing, among other things, an oxygen supply and propellant for thrusters. This module would be jettisoned before reentry. Much larger than a Mercury capsule, Gemini 3 was lifted into space by a Titan II, a two-stage rocket delivering 430,000lb (196,000kg) of thrust at liftoff.

The spacecraft was still small for two men to occupy on missions that would eventually last for weeks. Sitting side-by-side, they each had less room than you would find in a phone booth. But for pilots it was engagingly like the cockpit of a high-performance fighter, with ejection seats and a complicated set of controls designed to allow the pilots to change their orbit and eventually dock with another vehicle in space.

Not all the Gemini missions went smoothly. The object was to learn, and learning was often from mistakes. The most important goal of the program was to achieve docking between two spacecraft. However, the Gemini 6 mission to dock with an unmanned Agena vehicle was called off when the Agena broke up during launch. Another use was found for Gemini 6 when

it was sent up to rendezvous with Gemini 7 in December 1965. The craft maneuvered to within a few feet of one another, and Gemini 7's crew of Frank Borman and James Lovell went on to stay in space for two weeks. Four months later Gemini 8, crewed by Neil Armstrong and Dave Scott,



SPACE RENDEZVOUS

Astronauts Wally Schirra and Tom Stafford aboard Gemini 6 had this close-up view of Gemini 7 as they drew near during the first Gemini orbital rendezvous on December 15, 1965. This was a vital preparation for an eventual manned Moon mission.